# 10-701: Introduction to Machine Learning
# Lecture 5 – MLE & MAP

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# Front Matter

- Announcements:
  - 30 minute quiz this Friday at 2pm in Baker Hall A36 (our normal recitation time/place)
  - please submit your AWS credit request for the project, due Thursday, Jan 29 at 11:59pm
- Recommended Readings:
  - Mitchell, Estimating Probabilities

# Your first consulting job

- A billionaire from the suburbs of Seattle asks you a question:
  - He says: I have a coin, if I flip it, what's the probability it will fall with the head up?
  - You say: Please flip it a few times:

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  - He says: I have a coin, if I flip it, what's the probability it will fall with the head up?
  - You say: Please flip it a few times:

  - You say: The probability is: **3/5**
  - **He says: Why???**
  - You say: Because… frequency of heads in all flips

# Questions

- Why frequency of heads?

- How good is this estimation?
  - Would you be willing to bet money on your guess of the probability?
  - Why not?

# Model

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- What is the experimental data?
- Coin Flips

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# Model

- A model for coin flips
  - Bernoulli Distribution
- X is a random variable with Bernoulli distribution when:
  - X takes values in {0,1}
  - $P(X = 1) = p$
  - $P(X = 0) = 1 - p$
  - Where p in [0,1]

# Model

- X is a random variable with Bernoulli distribution when:
  - X takes values in {0,1}
  - P(X = 1) = p
  - P(X = 0) = 1 - p
  - Where p in [0,1]
- X = 1 i.e. heads with probability p, and X = 0 i.e. tails with probability 1 – p
  - Coin with probability of flipping heads = p
- And we draw **independent** samples that are **identically distributed** from same distribution
  - flip the same coin multiple times

# Bernoulli distribution

Data, D =


- P(Heads) = $\theta$, P(Tails) = $1-\theta$
- Flips are **i.i.d.**:
  - **Independent** events
  - **Identically distributed** according to Bernoulli distribution

<u>Choose $\theta$ that maximizes the probability of observed data</u>

# Probability of one coin flip

Let's say we observe a coin flip $X \in \{0, 1\}$.

The probability of this coin flip,
given a Bernoulli distribution with parameter $p$:

$$p^X (1-p)^{1-X}.$$

Equal to $p$ when $X = 1$, and equal to $(1-p)$ when $X = 0$.

# Probability of Multiple Coin Flips

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…Independence of samples

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$$

…probability of a Bernoulli sample

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&= \prod_{i=1}^{n} p^{X_i}\, (1-p)^{1-X_i} \\
&= p^{\sum_{i=1}^{n} X_i}\, (1-p)^{n-\sum_{i=1}^{n} X_i}
\end{aligned}
$$

$$\ldots\, p^a\, p^b = p^{a+b}$$

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&= p^{\sum_{i=1}^{n} X_i}\, (1-p)^{n-\sum_{i=1}^{n} X_i} \\
&= p^{n_h}\, (1-p)^{n-n_h}.
\end{aligned}
$$

where $n_h$ is the number of heads,
$n$ is the total number of coin flips

# Maximum Likelihood Estimator (MLE)

The MLE solution is then given by solving the following problem:

$$\begin{aligned}\widehat{p} &= \arg\max_{p} \mathbb{P}(X_1, \ldots, X_n; p) \\ &= \arg\max_{p} \left\{ p^{n_h} \left(1-p\right)^{n-n_h} \right\}\end{aligned}$$

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…argmax_x f(x) = argmax_x log f(x)

# MLE for coin flips

The MLE solution is then given by solving the following problem:

$$\widehat{p} = \arg\max_{p} \{n_h \log p + (n - n_h) \log(1 - p)\}$$

$$\implies \frac{n_h}{\widehat{p}} - \frac{n - n_h}{1 - \widehat{p}} = 0$$

$$\implies \widehat{p} = \frac{n_h}{n}.$$

# Maximum Likelihood Estimation

Choose θ that maximizes the probability of observed data

$$\widehat{\theta}_{MLE} = \arg\max_{\theta} \; P(D \mid \theta)$$

MLE of probability of head:

$$\widehat{\theta}_{MLE} = \frac{\alpha_H}{\alpha_H + \alpha_T}$$ **= 3/5**

"Frequency of heads"

# How many flips do I need?

$$\hat{\theta}_{MLE} = \frac{\alpha_H}{\alpha_H + \alpha_T}$$

- Billionaire says: I flipped 3 heads and 2 tails.
- You say: $\theta$ = 3/5, it is the MLE!
- He says: What if I flipped 30 heads and 20 tails?
- You say: Same answer, it is the MLE!
- **He says: If you get the same answer, would you prefer to flip 5 times or 50 times?**
- You say: Hmm… The more the merrier???
- He says: Is this why I am paying you the big bucks???

# KEY QUESTION: HOW GOOD IS THE MLE (OR ANY OTHER ESTIMATOR)?

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By the Law of Large Numbers!

…since the sample mean converges to
E(X) = p

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…linearity of expectation:
E(a X + b Y) = a E(X) + b E(Y)

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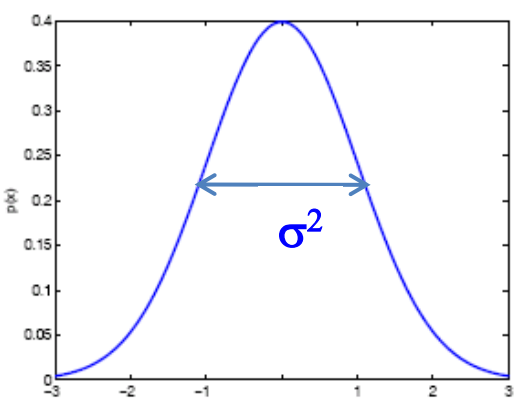
&= \mathbb{E}X_1 \\
&= p.
\end{aligned}
$$

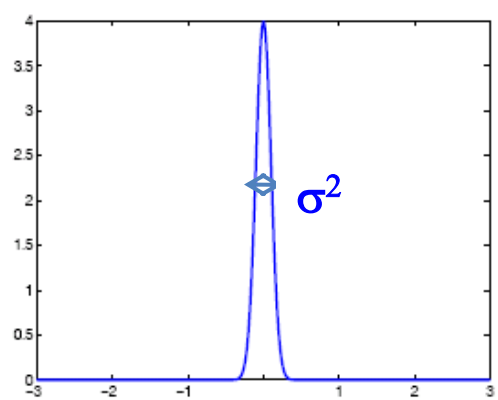
# What about continuous variables?

- Billionaire says: If I am measuring a continuous variable, what can you do for me?
- **You say: Let me tell you about Gaussians…**

$$P(x \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$ = N(μ,σ²)




μ=0

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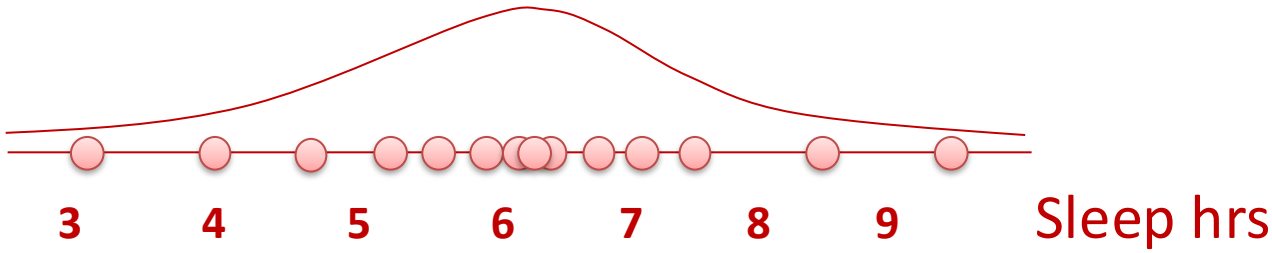
# Gaussian distribution

Data, D =




- Parameters: $\mu$ – mean, $\sigma^2$ - variance
- Sleep hrs are **i.i.d.**:
  - **Independent** events
  - **Identically distributed** according to Gaussian distribution

# MLE for Gaussian mean and variance

$$\widehat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\widehat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n} (x_i - \widehat{\mu})^2$$

Note: MLE for the variance of a Gaussian is **biased**

- Expected result of estimation is **not** true parameter!
- Unbiased variance estimator:

$$\widehat{\sigma}^2_{unbiased} = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \widehat{\mu})^2$$

# MLE for parametric models

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Model: $P(X; \theta)$ with parameters $\theta$.

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R. A. Fisher

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Likelihood Function: $L(\theta) := \prod_{i=1}^{n} P(X_i; \theta)$
The probability of seeing data $X_1, X_2, \ldots, X_n$ assuming parameters were $\theta$.

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The probability of seeing data $X_1, X_2, \ldots, X_n$ assuming parameters were $\theta$.

Maximum Likelihood Estimator (MLE): find that parameter $\theta$ that would maximize the likelihood of $\theta$

i.e. pick the $\theta$ that would maximize the probability of having seen the data that we do see

# Unbiasedness

An estimator $\widehat{\theta}(X_1, \ldots, X_n)$ where $X_i \sim P(X; \theta^*)$ is unbiased if

$$\mathbb{E}(\widehat{\theta}) = \theta^*.$$

MLE is ”asymptotically” unbiased i.e. there are some error terms that go to zero as a function of n, the number of samples

# Consistency

An estimator $\widehat{\theta}(X_1, \ldots, X_n)$ where $X_i \sim P(X; \theta^*)$ is consistent if $\widehat{\theta} \to \theta^*$ in probability as $n \to \infty$.

MLE is consistent under some mild regularity conditions on the model, and when the model size is fixed.

# How many flips?

- But recall the Billionaire's question:
  - How many flips would you prefer: 5 or 50?
  - How many flips would you need to be willing to bet money on your answer?
- Unbiasedness and Consistency do not answer this question
- We need convergence rates for our estimator
  - We will study more in Learning Theory lectures

# MLE

- Well-studied question in Statistics
  - Guarantees (consistency, unbiasedness, rates)
- What has Machine Learning contributed to this statistical question:
  - Specific kinds of guarantees e.g. sample complexity
  - New tools to derive guarantees (VC Dimension, etc.)
  - Computational Issues

# Computational Issues

- When number of parameters, or number of samples n is large, computing the MLE is a **large-scale optimization problem**
- Well-studied problem in optimization/operations research
- Machine Learning has contributed considerably via:
  - Better understanding of optimization problems that arise from statistical estimators such as MLE (in contrast to general optimization problems)

# Recall: Your first consulting job

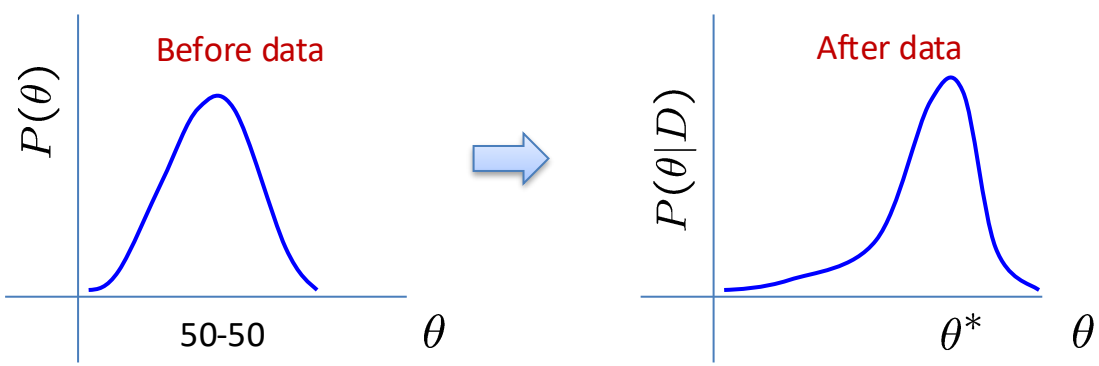
- A billionaire from the suburbs of Seattle asks you a question:
  - He says: I have a coin, if I flip it, what's the probability it will fall with the head up?
  - You say: Pl

  - You say: The probability is: **3/5** because… frequency of heads in all flips
  - **He says: But can I put money on this estimate?**
  - You say: ummm…. Maybe not.
    - Not enough flips (less than sample complexity)

# What about prior knowledge?

- Billionaire says: Wait, I know that the coin is "close" to 50-50. What can you do for me now?
- **You say: I can learn it the Bayesian way…**

- Rather than estimating a single θ, we obtain a distribution over possible values of θ



# Bayesian Learning

- Use Bayes rule:

likelihood prior

$$P(\theta \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid \theta) P(\theta)}{P(\mathcal{D})}$$

Parameters

Data

**Bayes, Thomas (1763)** An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London,* **53:370-418**

# Bayesian Learning

- Use Bayes rule:

$$P(\theta \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid \theta) P(\theta)}{P(\mathcal{D})}$$

- Or equivalently:

$$P(\theta \mid \mathcal{D}) \propto P(\mathcal{D} \mid \theta) P(\theta)$$

posterior likelihood prior

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# AIDS test (Bayes rule)

**Data**

- **Approximately 0.1% are infected**
- **Test detects all infections**
- **Test reports positive for 1% healthy people**

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**Data**

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- **Test reports positive for 1% healthy people**

Probability of having AIDS if test is positive:

$$P(a=1|t=1) = \frac{P(t=1|a=1)P(a=1)}{P(t=1)}$$

$$= \frac{P(t=1|a=1)P(a=1)}{P(t=1|a=1)P(a=1) + P(t=1|a=0)P(a=0)}$$

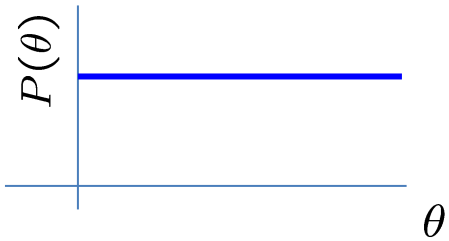
$$= \frac{1 \cdot 0.001}{1 \cdot 0.001 + 0.01 \cdot 0.999} = 0.091$$

Only 9%!...

# Prior distribution

- From where do we get the prior?
  - Represents expert knowledge (philosophical approach)
  - Simple posterior form (engineer's approach)

- Uninformative priors:
  - Uniform distribution




- Conjugate priors:
  - Closed-form representation of posterior
  - $P(\theta)$ and $P(\theta|D)$ have the same algebraic form as a function of $\theta$

# Conjugate Prior

- P(θ) and P(θ|D) have the same form as a function of theta

Eg. 1 Coin flip problem

Likelihood given Bernoulli model:

$$P(\mathcal{D} \mid \theta) = \theta^{\alpha_H}(1-\theta)^{\alpha_T}$$

If prior is Beta distribution,

$$P(\theta) = \frac{\theta^{\beta_H - 1}(1-\theta)^{\beta_T - 1}}{B(\beta_H, \beta_T)} \sim Beta(\beta_H, \beta_T)$$

Then posterior is Beta distribution

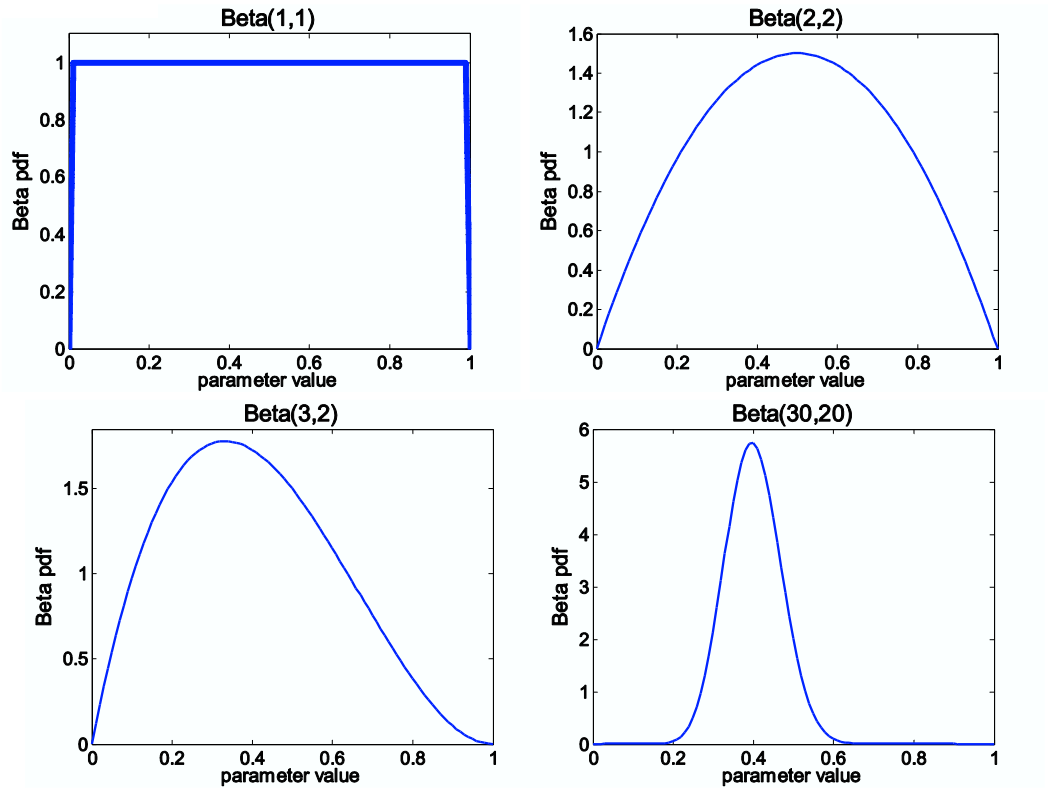
$$P(\theta|D) \sim Beta(\beta_H + \alpha_H, \beta_T + \alpha_T)$$

**For Binomial, conjugate prior is Beta distribution.**

# Beta distribution

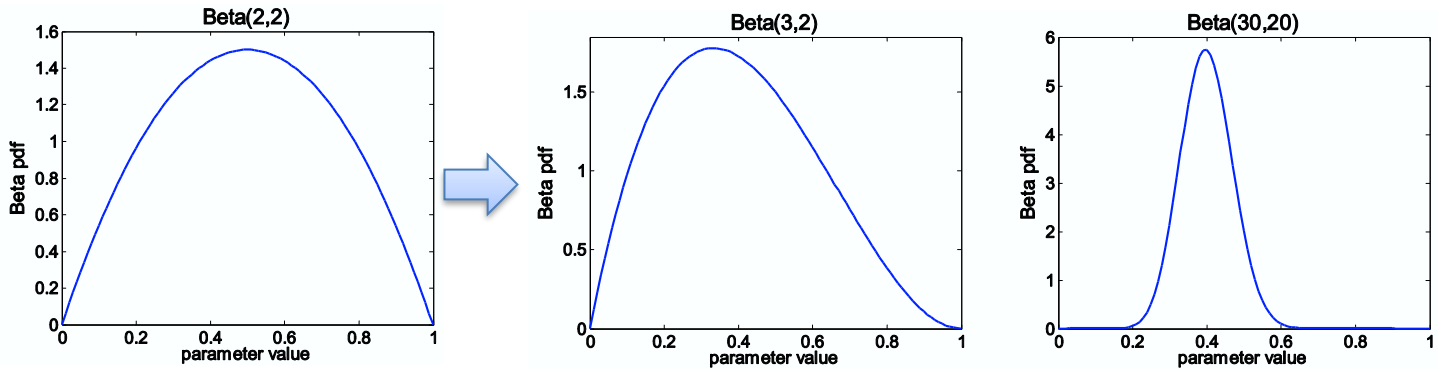
$Beta(\beta_H, \beta_T)$

More concentrated as values of $\beta_H$, $\beta_T$ increase



# Beta conjugate prior

$$P(\theta) \sim Beta(\beta_H, \beta_T) \qquad P(\theta|D) \sim Beta(\beta_H + \alpha_H, \beta_T + \alpha_T)$$




As $n = \alpha_H + \alpha_T$ increases

As we get more samples, effect of prior is "washed out"

# Conjugate Prior

- P(θ) and P(θ|D) have the same form

Eg. 2 Dice roll problem (6 outcomes instead of 2)

Likelihood is ~ Multinomial($\theta = \{\theta_1, \theta_2, \ldots, \theta_k\}$)

$$P(\mathcal{D} \mid \theta) = \theta_1^{\alpha_1}\theta_2^{\alpha_2}\ldots\theta_k^{\alpha_k}$$

If prior is Dirichlet distribution,

$$P(\theta) = \frac{\prod_{i=1}^{k}\theta_i^{\beta_i - 1}}{B(\beta_1, \ldots, \beta_k)} \sim \text{Dirichlet}(\beta_1, \ldots, \beta_k)$$

Then posterior is Dirichlet distribution

$$P(\theta|D) \sim \text{Dirichlet}(\beta_1 + \alpha_1, \ldots, \beta_k + \alpha_k)$$

**For Multinomial, conjugate prior is Dirichlet distribution.**

# Posterior Distribution

- The approach seen so far is what is known as a **Bayesian** approach
- Prior information encoded as a **distribution** over possible values of parameter
- Using the Bayes rule, you get an updated **posterior** distribution over parameters, which you provide with flourish to the Billionaire
- But the billionaire is not impressed
  - Distribution? I just asked for one number: is it 3/5, 1/2, what is it?
  - How do we go from a distribution over parameters, to a single estimate of the true parameters?

# Maximum A Posteriori Estimation

Choose θ that maximizes a posterior probability

$$\begin{aligned}\widehat{\theta}_{MAP} &= \arg\max_{\theta} \quad P(\theta \mid D) \\ &= \arg\max_{\theta} \quad P(D \mid \theta) P(\theta)\end{aligned}$$

$$P(\theta|D) \sim Beta(\beta_H + \alpha_H, \beta_T + \alpha_T)$$

MAP estimate of probability of head:

$$\widehat{\theta}_{MAP} = \frac{\alpha_H + \beta_H - 1}{\alpha_H + \beta_H + \alpha_T + \beta_T - 2}$$

Mode of Beta distribution

# MLE vs. MAP

- Maximum Likelihood estimation (MLE)

  Choose value that maximizes the probability of observed data

$$\widehat{\theta}_{MLE} = \arg\max_{\theta} P(D|\theta)$$

- Maximum *a posteriori* (MAP) estimation

  Choose value that is most probable given observed data and prior belief

$$\begin{aligned}\widehat{\theta}_{MAP} &= \arg\max_{\theta} P(\theta|D) \\ &= \arg\max_{\theta} P(D|\theta)P(\theta)\end{aligned}$$

When is MAP same as MLE?

# MLE vs. MAP

$$\hat{\theta}_{MLE} = \frac{\alpha_H}{\alpha_H + \alpha_T}$$

What if we toss the coin too few times?

- You say: Probability next toss is a head = 0
- Billionaire says: You're fired! …with prob 1 ☺

$$\hat{\theta}_{MAP} = \frac{\alpha_H + \beta_H - 1}{\alpha_H + \beta_H + \alpha_T + \beta_T - 2}$$

- Beta prior equivalent to extra coin flips
- As *n* → infty, prior is "forgotten"
- **But, for small sample size, prior is important!**

# MLE vs MAP

You are no good when sample is small
You give a different answer for different priors

# MAP for Gaussian mean and variance

- Conjugate priors
  - Mean: Gaussian prior
  - Variance: Wishart Distribution

- Prior for mean:

$$P(\mu \mid \eta, \lambda) = \frac{1}{\lambda\sqrt{2\pi}} e^{\frac{-(\mu-\eta)^2}{2\lambda^2}}$$ = N(η,λ²)

# MAP for Gaussian Mean

$$\widehat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\widehat{\mu}_{MAP} = \frac{\frac{1}{\sigma^2}\sum_{i=1}^{n} x_i + \frac{\eta}{\lambda^2}}{\frac{n}{\sigma^2} + \frac{1}{\lambda^2}}$$

# Prior Information

- In the Bayesian approach, the prior information is encoded through a prior distribution over the parameters
- Seems onerous: the distribution typically seems to be obtained from convenience (conjugate distribution)
- What other ways can we encode our prior knowledge about the parameters?
- A non-Bayesian approach is via constraints

# Encoding prior information via constraints

MLE:

$$\max_{\theta} \log \mathbb{P}(D; \theta).$$

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Constrained MLE:

$$\max_{\theta} \log \mathbb{P}(D; \theta)$$
$$\text{s.t.}\, \mathcal{R}(\theta) <= C.$$

# Encoding prior information via constraints

MLE:

$$\max_{\theta} \log \mathbb{P}(D; \theta).$$

Constrained MLE:

$$\begin{aligned} &\max_{\theta} \log \mathbb{P}(D; \theta) \\ &\text{s.t.}\, \mathcal{R}(\theta) <= C. \end{aligned}$$

When $\mathcal{R}(\theta)$ is convex, constrained MLE is equivalent to regularized MLE:

$$\max_{\theta} \left\{ \log \mathbb{P}(D; \theta) + \lambda\, \mathcal{R}(\theta) \right\}.$$

# Regularized MLE

Regularized MLE:

$$\max_{\theta} \left\{ \log \mathbb{P}(D; \theta) + \lambda \, \mathcal{R}(\theta) \right\}.$$

Trades off maximizing the log-likelihood (i.e. fit to data), against the "prior" constraints encoded by regularization (which do not involve the data at all).

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The MAP estimator can be seen to be a special case by simply setting

$$\lambda \, \mathcal{R}(\theta) = \log \, P(\theta).$$

Here, the tradeoff between likelihood and prior is naturally captured by setting the regularization function equal to the log of the prior distribution.

# Popular Regularization functions

- $\ell_2$ **regularization**:

$$\mathcal{R}(\theta) = \|\theta\|_2^2 = \sum_{j=1}^{p} \theta_j^2.$$

This regularization encodes the prior information that the parameter values are not too large (where how large is determined by the regularization tradeoff parameter $\lambda$).

This regularization is thus a "general purpose" regularization function (who wants their parameters to be very large?)

# Popular Regularization functions

- $\ell_1$ **regularization**:

$$\mathcal{R}(\theta) = \|\theta\|_1 = \sum_{j=1}^{p} |\theta_j|.$$

This regularization encodes the prior information that the parameter values be **sparse**: i.e. with many zero values.

This is a very important prior constraint in big data settings:
with very large number of parameters, we expect the true model to depend on only a few non-zero parameters.

Widely used in high-dimensional model learning: called LASSO when used with linear regression models.

POLL: https://forms.gle/b9ajn8d5vyM5T6P46

Scan me

# Poll: When does MAP equal MLE?

Suppose we estimate a parameter $\theta$ using MAP:

$\theta_MAP = \text{argmax}_\theta\ P(D \mid \theta)\ P(\theta)$.

For which choice of prior $P(\theta)$ will $\theta_MAP$ be identical to $\theta_MLE$ for any dataset D?

A. A uniform (constant) prior over $\theta$

B. A symmetric Beta($\beta$,$\beta$) prior with $\beta > 1$

C. A prior highly concentrated near $\theta = 0.5$

D. Any prior, as long as the sample size is small

E. Any prior; MAP always equals MLE